

How Many Molecules Are in a Yeast Cell, and Who Has Already Counted Them

A scoping note on retrobiosynthetic gap filling, and what is left to do

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Abstract

A full physical accounting of a yeast cell's chemical species would let us represent every metabolite, medium component and dosed inhibitor in one molecular space, and the genome-scale model is the obvious starting inventory. Yeast9 names 1,378 distinct species of which 894 carry a structure, but only 273 of those structures name a single molecule rather than a set of stereoisomers, and a retrobiosynthetic expansion published in 2026 has already raised yeast metabolome coverage from 7 to 92 percent using 21,921 reaction rules. Computing the dry-mass pools from the model's own biomass equation closes the accounting to 96 percent and puts DNA at 0.39 percent and the cofactor and ion pools at 0.73 percent, so the chemical species that matter here are numerous and under one percent of the cell by mass. The opening that remains is therefore not the expansion but two things nobody reports: stereochemical resolution, and the lipid classes no reconstruction structures.

1 The question x

Experiment 031 represents a dosed inhibitor, a medium component and a metabolic-model metabolite in one embedding space, which works only to the extent that the molecules are actually named and structured. That raised a broader question worth scoping before it is acted on: how many distinct chemical species does a yeast cell contain, how many does the genome-scale reconstruction name, and could a retrobiosynthetic method enumerate the rest.

The motivation is a physical accounting rather than a modeling convenience. If every species carried a resolved structure with an atom mapping, then reaction rules, thermodynamics and molecular embeddings would all read the same object, and gap filling would become a question about chemistry rather than about identifiers. The practical target is the set of intermediates reachable from a defined medium, including the uncatalyzed and spontaneous chemistry that constraint-based reconstruction tends to leave out.

This note measures what the reconstruction gives us, reports what the literature has already settled, and states what is left. It is short on purpose: the point is to decide whether the side quest is worth starting.

2 What the reconstruction actually names x

Figure 1 measures yeast-GEM 9.0.2 as an inventory of chemical species rather than as a flux model.

Four measurements matter.

The inventory is smaller than the model's own headline number. The release reports 2,805 metabolites, but those are compartment-specific entries. Collapsed to distinct chemical species the model names **1,378**, spread over 14 compartments as 2,806 entries in the file we read. Of those species 894 carry a SMILES and 484 do not.

Only a fifth of the model is pinned to a single molecule. Of the 894 structures, **621** carry at least one unassigned stereocenter, so the string names a set of stereoisomers rather than a compound. Only **273 of 1,378** species, a fifth of the model, are specified precisely enough for a physical accounting to treat them as one species. This is the measurement that bears most directly on whether an atom-mapped treatment is possible, and it is not reported anywhere in the literature we mirror.

Half the reactions are structurally complete. Every participant carries a structure in **2,156 of 4,131** reactions, 52 percent. That is the share a structure-aware or rule-based method could read without first resolving something, and it is the practical ceiling on applying reaction rules to this model as released.

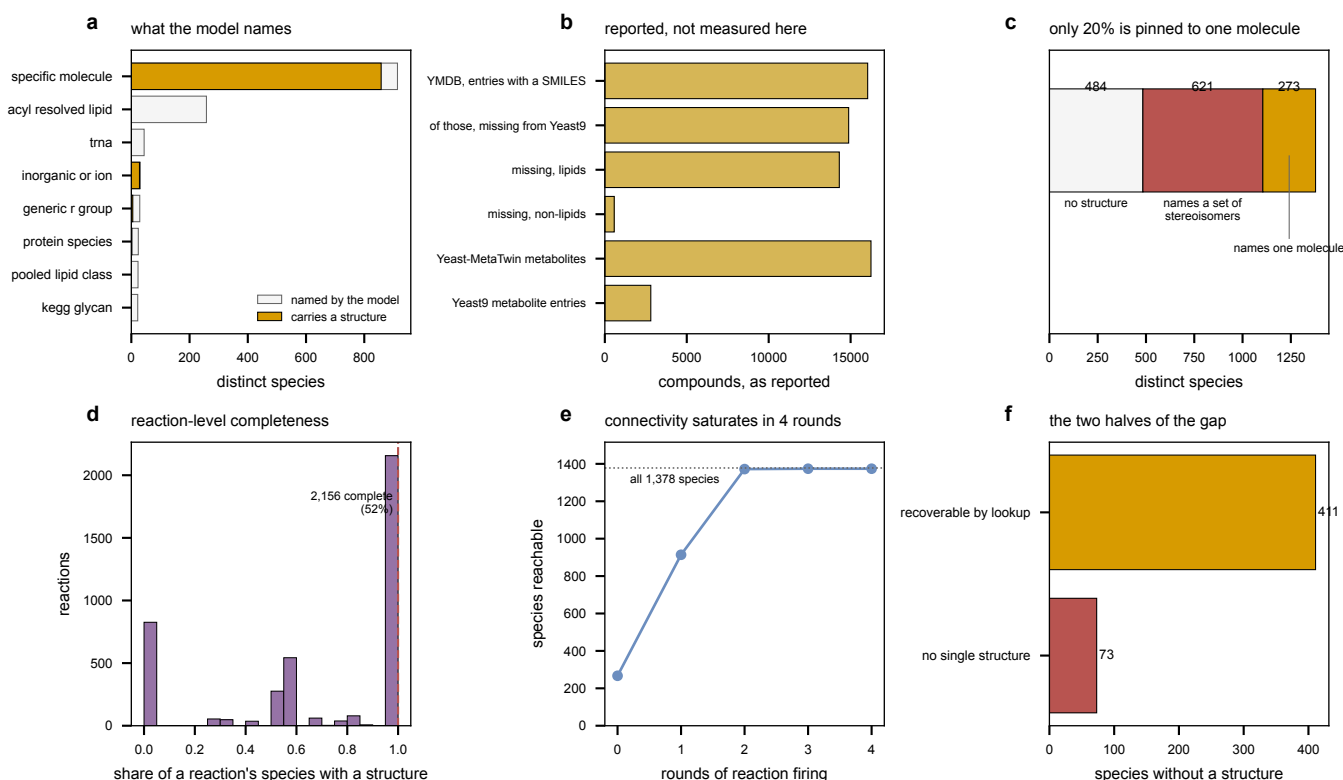


Figure 1: Yeast9 names 1,378 chemical species, and only a fifth of them are pinned to a single molecule. The reconstruction read as a chemical inventory rather than as a flux model. **a**, Distinct species by structure class, with the share carrying a SMILES. **b**, Published metabolome counts, which this script does not measure. **c**, How precisely a species is named: no structure, a structure with an unassigned stereocenter that names a set of stereoisomers, or a structure that names one molecule. **d**, Share of each reaction's participants carrying a structure. **e**, Species reachable from the model's own boundary metabolites when a reaction fires as soon as any one substrate is present, an upper bound that ignores stoichiometry and thermodynamics and that saturates in four rounds. **f**, The species without a structure, split into those a lookup could recover and those that can never have one.

Connectivity gives no useful bound. Starting from the model’s 267 boundary metabolites and firing a reaction whenever any one of its substrates is present, **1,374 of 1,378 species are reached within four rounds**. The rule is deliberately generous, and the result is that graph connectivity says only that the network is dense. It follows that bounding the space of intermediates needs real reaction rules with atom mapping rather than reachability on a species graph, which is an argument for the retrobiosynthetic framing rather than against it.

One number to treat carefully. Counting metabolite entries that participate in a single non-boundary reaction or that can only be produced or only consumed gives **747 of 2,806** on 9.0.2. The published counterpart for Yeast8 is 464 of 2,742 [▲](#) [external: Chen, Li and Nielsen 2022]. The two are not directly comparable because the definitions of a dead end may differ; ours is stated in the script and is the more inclusive reading.

3 The expansion has already been done ✗

The scoping question has a published answer, and it arrived in 2026. Yeast-MetaTwin [▲](#) [external: Wu et al. 2026, “Systematically exploring yeast metabolism through retrobiosynthesis and deep learning”; Nielsen and Kerkhoven are authors] applies exactly the method under consideration here, at a scale that would be hard to improve on casually.

The numbers, quoted through `yeast9\_literature\_counts.py` and plotted hatched in Figure 2. The Yeast Metabolome Database holds **16,042** entries carrying a SMILES, of which **14,882 are absent from Yeast9**, and those missing entries are **14,310 lipids against 572 non-lipids**. The authors extracted **21,921 enzyme-associated reaction rules** and **213 spontaneous rules** from MetaNetX and MetaCyc, report that 95 percent of the rules regenerate their original reactions, and trimmed the resulting pool to a connected network of **1,092,946 reactions**. The outcome they state is that coverage of the yeast metabolome rose **from 7 percent to 92 percent**, and for the non-lipid metabolome from 54 to 75 percent.

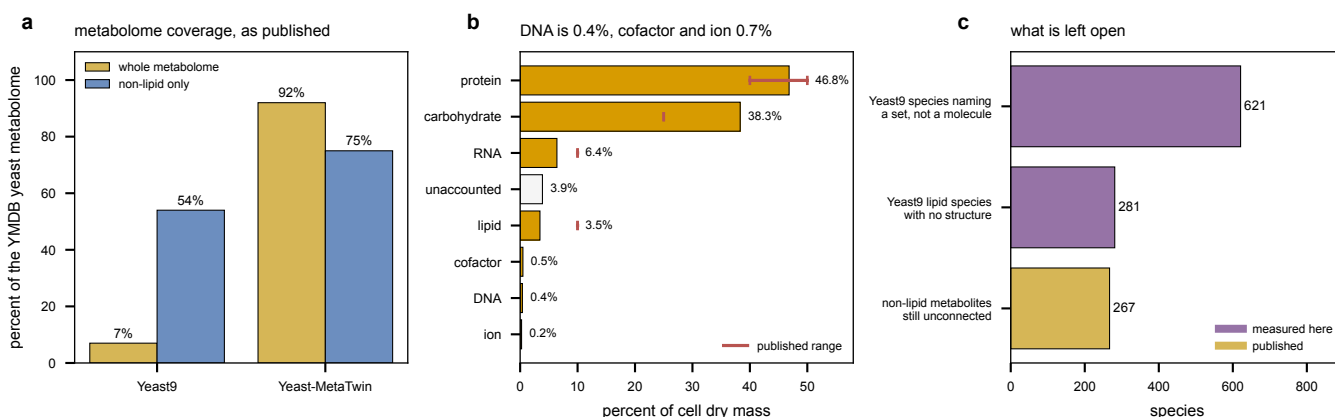


Figure 2: The retrobiosynthetic expansion is published, and what is left open is stereochemistry, lipids and a missing mass fraction. Hatched bars are published and were not measured here; solid bars are measured from the release. **a**, Metabolome coverage before and after the published expansion, for the whole metabolome and for the non-lipid subset. **b**, Yeast dry-mass composition computed from the model’s own biomass pseudo-reactions, which closes to 96.1 percent; red whiskers are the published range for the four pools the literature covers. DNA is 0.39 percent and the cofactor and ion pools together 0.73 percent, so the remainder after the four large pools is not a five-to-ten percent small-molecule fraction. **c**, What remains: Yeast9 species whose structure names a set rather than a molecule, Yeast9 lipid species with no structure at all, and the non-lipid metabolites the published expansion could not connect.

Two consequences follow immediately. First, proposing a retrobiosynthetic enumeration of yeast metabolism as new work would duplicate a published result from the same group that maintains the reconstruction. Second, the residual they report is specific and small: **267 non-lipid metabolites**, 1.7 percent of the metabolome, could not be connected without introducing metabolites outside the yeast metabolome.

The mass question, answered from the model’s own biomass equation. The rounded figures in the literature do not add up and say nothing about DNA or the small-molecule pool: yeast is reported as 40 to 50 percent protein, about 10 percent RNA, about 10 percent lipid and about 25 percent cell wall [▲](#) [external: Milo and Phillips 2016; Feldmann 2012]. The reconstruction carries a complete answer that needs no lookup. Its biomass pseudo-reaction draws one unit each from a protein, carbohydrate, RNA, DNA, lipid, cofactor and ion pool, and every pool lists its constituents with coefficients in millimoles per gram dry weight, so multiplying by formula mass gives grams per gram directly (Figure 2b).

Measured that way the pools are **46.8 percent protein**, **38.3 percent carbohydrate**, **6.4 percent RNA** and **3.5 percent lipid**, with **0.48 percent cofactor**, **0.39 percent DNA** and **0.25 percent ion**. They sum to 96.1

percent, and the 3.9 percent shortfall is the few constituents carrying no formula, so the accounting nearly closes and the residual is named rather than left implicit.

That settles the question the rounded figures leave open, and the answer is not what the arithmetic suggests. **The remainder after protein, carbohydrate, RNA and lipid is not five to ten percent of small molecules.** DNA is 0.39 percent and the cofactor and ion pools together are 0.73 percent. Two caveats on reading those numbers. The model's carbohydrate pool runs above the 25 percent cell-wall figure because it also carries storage glycogen and trehalose, and its lipid pool runs below the published 10 percent. And the cofactor and ion pools are a **lower bound** on the small-molecule pool, because they hold the cofactors and ions the biomass equation requires rather than the cell's whole metabolite pool; a soluble pool measured by extraction would be larger, and the *E. coli* figure for that quantity is 3 to 3.9 percent of dry weight.

The practical consequence for a chemical accounting is that the pool it is about is small. If the target is every distinct chemical species, the species are numerous and they are under one percent of the cell by mass, so the exercise is about identity and connectivity rather than about mass balance.

4 What is left, and the question for a collaborator ✗

Three openings survive the published work, and they are narrower and more specific than the original framing.

Stereochemical resolution, which nobody reports. A rule-based expansion operates on SMILES, and 621 of the 894 Yeast9 structures name a set of stereoisomers rather than a molecule. A species that is stereochemically ambiguous cannot carry an atom mapping that means anything, cannot be assigned a unique thermodynamic value, and collapses L and D forms onto one key. Resolving the 1,378 species to single molecules is a prerequisite for a physical accounting, it is measured here as a gap, and no paper in the mirror reports it for any yeast reconstruction. This is the cleanest opening.

Lipids, where both the model and the expansion are weakest. Lipids dominate every count. Of Yeast9's 1,378 species, **281 are acyl-resolved or pooled lipid classes and not one carries a structure.** Of the 14,882 metabolites the published comparison found missing, 14,310 are lipids. And the published expansion reaches 92 percent of the whole metabolome but only 75 percent of the non-lipid subset, which means the whole-metabolome figure is carried by lipids enumerated combinatorially from acyl chains rather than by resolved chemistry. Whether that enumeration is a physical accounting or a parameterization is a fair question to put to a specialist.

The mass fraction, which is simply missing. Nothing in the mirrored literature states what share of yeast dry mass is small molecules. Until that number exists the exercise has no denominator: an accounting of every chemical species is a different undertaking if the pool is 3 percent of the cell than if it is 15.

Does retrobiosynthesis beat constraint-based gap filling here. Worth stating precisely, because the two do different jobs. Constraint-based gap filling adds reactions from a universal database until a model carries flux to a required biomass component, so it is driven by a phenotype and answers with the smallest set of reactions that restores it. It never proposes a molecule that is not already in its database. A retrobiosynthetic expansion applies reaction rules to structures and generates species that were in no database, so it can propose the intermediate itself. For enumerating the space of chemical intermediates, which is the stated goal, only the second can work. The measurement in Section 2 supports that from the other direction: reachability on the species graph saturates at 99.7 percent in four rounds and therefore bounds nothing, so the bound has to come from chemistry.

Hypothesis (untested): the residual the published expansion reports, 267 non-lipid metabolites that could not be connected with enzyme-annotated reactions, would shrink if uncatalyzed chemistry were admitted on the same footing as enzymatic rules, since spontaneous and metal-catalyzed steps are exactly what an enzyme-annotated rule set excludes. The published work carries 213 spontaneous rules against 21,921 enzymatic ones, which is a small allowance for a cell held at 30 degrees for many hours.

The concrete ask. Put to a retrobiosynthesis specialist, the question is not whether yeast metabolism can be expanded, because it has been. It is three narrower things. Can a rule set operate on stereochemically resolved species, and what does it cost to resolve 1,378 of them first. Does admitting uncatalyzed and metal-catalyzed chemistry close the 267-metabolite residual, or does it inflate the space without bound. And is there a representation of acyl-resolved lipids that is a physical accounting rather than a combinatorial enumeration.

The deliverable that would make the conversation concrete is a conversion of Yeast9 into a reaction format a retrobiosynthesis tool consumes, which means per-reaction substrate and product structures with an atom mapping where one can be derived. Section 2 measures the ceiling on that conversion as released: **2,156 of 4,131 reactions**, 52 percent, have a structure for every participant today. That conversion is a small piece of work and it is the natural first step, but it is a side quest and it does not block the modeling phase of experiment 031.